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PALEOMAGNETISM AND U–Pb GEOCHRONOLOGY OF VOLCANIC ROCKS FROM MICHIPICOTEN ISLAND, LAKE SUPERIOR, CANADA: PRECISE CALIBRATION OF THE KEWEENAWAN POLAR WANDER TRACK

H.C. PALMER

Department of Geophysics, The University of Western Ontario, London, Ontario N6A 5B7 (Canada)

D.W. DAVIS

Geology Department, University of Toronto, Toronto, Ontario M5S 1A1 (Canada)

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Abstract

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Paleomagnetic poles have been determined for normally magnetized samples of late Proterozoic volcanic rocks from 23 sites on Michipicoten Island, Lake Superior, Canada. The results are in general accord with an earlier, less precise, study by one of the authors and indicate a small polar shift between rocks of the Mamainse Point Formation (186°E, 38°N) and the Michipicoten Island Formation (175°E, 26°N). A quartz-feldspar porphyry intruding the lower Mamainse Point Formation but unconformably overlain by the Michipicoten Island Formation was sampled for U–Pb zircon dating. The zircons define an age of emplacement of $1086.5 + 1.3/-3.0$ Ma. This age and the pole positions are consistent with, and supplement ages previously determined for the Keweenawan paleomagnetic polar wander curve. These data define an apparent polar wander rate of 14 cm a^{-1}. This age, combined with previously published U–Pb ages, indicates a time span of ~ 22 Ma for most Keweenawan igneous activity. In addition, the time of the Keweenawan paleomagnetic reversal is constrained to be between $1086.5 + 1.3/-3.0$ and 1097.6 ± 3.6 Ma, the age of the upper part of the reversely magnetized Osler Group. The zircon population contained an inherited component from a source 1403 ± 50 Ma old. This suggests that the porphyry magma was at least partly derived by melting of an anorogenic pluton emplaced during the mid-Proterozoic.

Introduction

The most temporally continuous and internally consistent part of the North American Precambrian paleomagnetic record occurs in the 900 Ma–1200 Ma interval (Roy, 1983). Most of the data constituting the older part of this record are derived from Keweenawan rocks of the Lake Superior Basin, whereas rocks from the Greenville Province have yielded the data for the younger portion. The details of the Keweenawan record have recently been reviewed by Halls and Pesonen (1982), and they show that the form of the record has changed little since it was first outlined by Dubois (1962). How-

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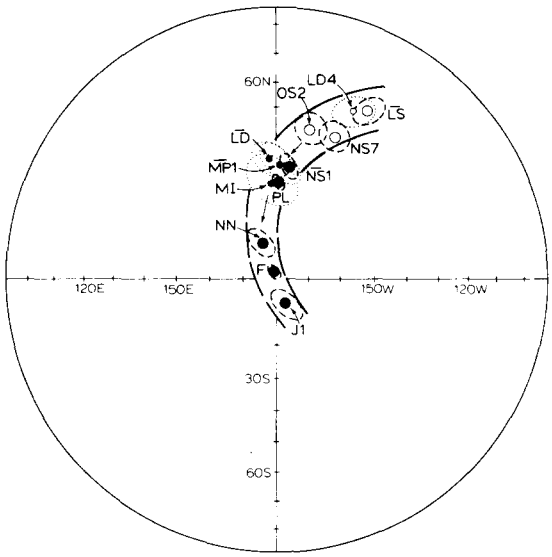


Fig. 1. Keweenawan paleomagnetic poles from Keweenawan units whose stratigraphic order can be inferred from geological field relations, magnetic anomaly correlation and radiometric age data (large symbols). The four poles indicated with smaller symbols are judged to be as reliable as those forming the control sequence but their stratigraphic position within the control sequence is less certain (Halls and Pesonen, 1982). Open/closed symbols represent poles of reversed/normal polarity. The poles plotted are as follows and more detailed information can be found in Halls and Pesonen, 1982, Appendix 1: LS, Logan Sills; LD4, Logan Dikes; NS7, lower North Shore Volcanics, OS2, upper part of Osler reversed sequence; NS1, mean of upper North Shore Volcanic Group poles; LD, mean Logan dikes; MP1, Mamainse Point upper normal sequence; MI, Volcanics of Michipicoten Island as given by Palmer, 1970; PL, Portage Lake Volcanic Group; NN, Nonesuch Shale; F, Freda Sandstone; J1, Jacobsville Sandstone.

ever, in the 20 year interval, more data have been added, less reliable data can be objectively filtered out, and sharper time constraints can be placed on the duration of the track (van Schmus et al., 1982). Specifically, beginning at ~ 1110 Ma during Logan Sill and Early Osler Group time (Davis and Sutcliffe, 1985) and ending at Nonesuch Shale time (1050 ± 50 Ma, Chaudhuri and Faure, 1967), the apparent polar path traversed about 60° of arc moving from the northeast Pacific to the central equatorial Pacific (Fig. 1). Not all of this apparent polar motion may reflect continental or plate displacement, however, as the northernmost poles are all of reverse polarity whereas poles at mid and low latitudes are of normal polarity. Some 20° of the polar displacement accompanies one and possibly two reverse-to- normal transitions and may well reflect a nongeocentric dipole source (Halls and Pesonen, 1982; Nevanlinna and Pesonen, 1983). Nevertheless, the north–south progression of the pole with time is consistent with stratigraphic relationships, cutting relationships and available geochronology. One possible exception within the normal polarity portion of the record occurs when the paleopoles for the Nonesuch Shale and Michipicoten Island rocks are compared. The Nonesuch pole lies to the south of the Michipicoten Island pole suggesting a younger magnetization age for the Nonesuch Shale. The Rb–Sr whole-rock ages for these units were originally reported by Chaudhuri and Faure (1967) and Baragar (1978) and have been corrected for the decay constants of Steiger and Jäger (1977). Errors quoted for these ages are those given by the original authors. These ages suggest an interchanged time sequence. The Nonesuch Shale isochron gives a 1050 ± 50 Ma age, whereas the first intrusive phase porphyries and the overlying Channel Lake Andesite, both from Michipicoten Island, give ages of 1004 ± 64 Ma and 918 ± 78 Ma. A more precise age for the Michipicoten Island rocks is clearly desirable. With it in hand, the possible necessity to postulate back-tracking along the polar track can be more definitely evaluated.

Part of the apparent discrepancy of the relative positions of the Nonesuch and Michipicoten Island poles may result from inaccuracies in the paleomagnetic data rather than from the imprecision of the age data. Although the paleomagnetic result from the Nonesuch Shale (Henry et al., 1977) was obtained by modern analytical methods, the Michipicoten Island result (Palmer, 1970) was not. In that early paleomagnetic study, much of the effort was directed towards establishing the polarity sequence in diverse Keweenawan sequences. The sampling scheme was designed to search for polarity zones, and for efficiency, typically one and at most two samples were collected at

each site. This sampling scheme limits the ability to filter out inconsistent within-site data. Since all of the rocks on Michipicoten Island were found to be of normal polarity, limited alternating field demagnetization was performed and stringent criteria for defining stable end points were not used. Furthermore, at the time of sampling for the 1970 study, an adequate geological map did not exist for the island. As a consequence, the stratigraphic relations and more importantly, the structural attitudes and their variations along strike, were only imperfectly known. Subsequent to this early paleomagnetic work, a modern geological study, together with a geological map at 1:50 000 scale, has been published (Annells, 1974). A re-examination, using modern demagnetization techniques and data analysis, of virgin samples still remaining from the 1970 study hinted (Raleigh, 1984) that a small directional shift may be recorded within the sequence on Michipicoten Island.

Accordingly, the rocks on Michipicoten Island were re-sampled during the summers of 1984 and 1985 for both paleomagnetic and age dating studies. The purposes of the present study are: (1) to document whether the 1970 pole is essentially correct and that it lies north of the Nonesuch Shale pole on the polar track; (2) to discover whether a polar shift is recorded within the sequence on the island, possibly coincident with the unconformity separating the Mamainse Point Formation and the overlying Michipicoten Island Formation; and (3) to test whether the time sequence of paleomagnetic poles, calibrated by high precision U–Pb dating for three reverse polarity Keweenawan units (Davis and Sutcliffe, 1985), can be extended to the normal polarity units. With these latter data in hand, a limit in time, can be placed on the polar shift accompanying the reverse to normal polarity transition widely recorded in Keweenawan rocks.

Geology and field methods

Michipicoten Island consists of a 4300 m thickness of basic, intermediate, and acid flows with associated clastic horizons; also included in this thickness are a number of broadly concordant intrusions of fine-grained basic and acid material (Annells, 1974). The volcanics have a southerly dip (Fig. 2) which decreases from 45°–50° on the north shore to 10°–15° in the uppermost flows on the south shore. Neither the top nor the base of the succession is exposed on the island. A 850 m sequence of thin, olivine-bearing basalt flows, correlated with the Mamainse Point Formation of eastern Lake Superior (Annells, 1974), forms the lowest exposed formation. The Mamainse Point Formation is invaded by a variety of sub-volcanic intrusions ranging from acid quartz porphyry to basaltic andesite. These rocks are overlain unconformably by the Michipicoten Island Formation. This latter unit has been subdivided into five major members and six minor members. The minor members are primarily agglomerates, tuffs and volcaniclastic sediments; they were not sampled in the present study. The three lowest of the major members, the Cuesta, the Channel Lake and the Quebec Harbour are each composed of one or two thick, glassy andesite flows. These are succeeded by the South Shore Member which consists of thinner tholeiitic basalt lava flows. The highest member of the Michipicoten Island Formation is the Davieaux Island Member which consists of a single pink-purple rhyolite lava. It outcrops only on a chain of islands parallel to the south coast and its contact with the South Shore Member is not exposed. The contact between these two members may be a fault. The cumulative thickness of the five major members of the Michipicoten Island Formation is in excess of 1300 m.

Distinguishing features of the Michipicoten Island rocks, as compared to other Keweenawan volcanic sequences, are the abundance of rocks of intermediate and acid composition and the very low grade of metamorphism. Although olivines are pseudomorphed by serpentine and chlorite in basalts of the Mamainse Point Formation and zeolites and silica minerals infill the vesicles in them, the degree of secondary mineral development is less than in most other Keweenawan basalt sequences exposed in the Lake Superior basin. The andesites and basalts of the Michipicoten Island Formation are extremely fresh; original volcanic glass remains undevitrified. The lack of evidence of thermally induced burial metamorphism is reflected in the paleomagnetic results (below) in that the natural remanence of the bulk of these rocks consists of a single magnetic component. However, the SiO_2 contents of the glassy mesostasis varies between 62 and 72% (Annells, 1974). As a consequence of the siliceous nature of the original melts, leading to high viscosities during eruption and flow, upper and lower flow contacts are not perfectly planar. For this reason, estimates of the paleohorizontal at some paleomagnetic sites within the Michipicoten Island Formation have uncertainties of 5°–10°.

Four to seven field-drilled cores (samples), oriented using both a magnetic and a sun compass, were taken at 20 sites. At the remaining five sites, oriented block samples, or a combination of block samples and field-drilled samples, were taken. Six block samples from the Mamainse Point Formation, distributed between sites 8 and 6 (Fig. 2) of the present

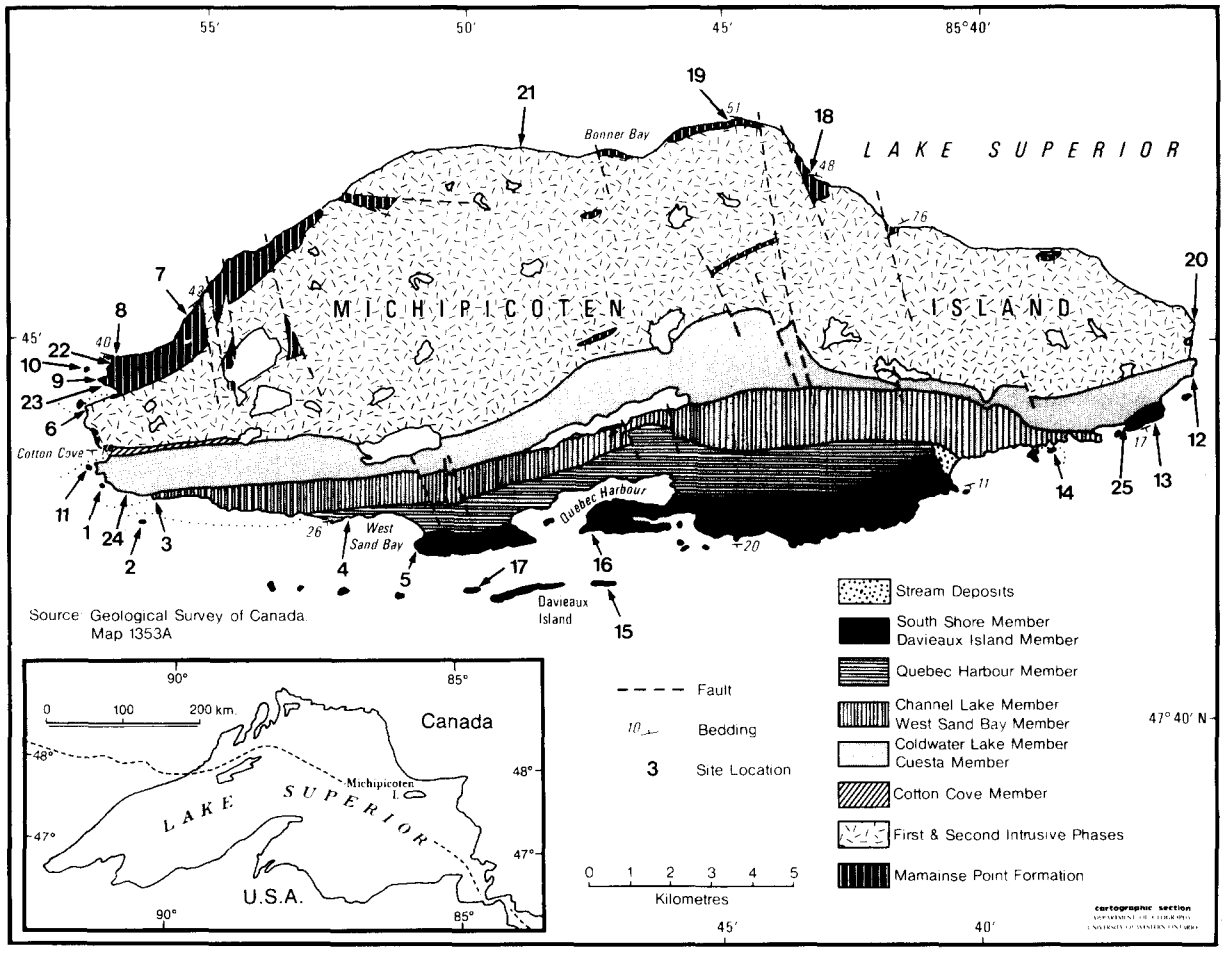


Fig. 2. Geological map of Michipicoten Island, generalized from Annells (1974) showing sampling sites.

study, were recovered from the 1970 collection (Palmer, 1970). Virgin specimens from these samples were studied by Raleigh (1984) using current laboratory and data analysis methods (see next section). Although several of these block samples probably come from sampling sites of the present study, others clearly do not. Inclusion of these results helps to average secular variation within the Mamainse Point Formation and for this reason these results have been given unit site weight in Table I where they have the site designation 'KM'.

Paleomagnetic techniques

From each oriented sample, at least one standard paleomagnetic specimen was prepared. The natural remanence of these samples was measured using a Schonstedt DSM-1 spinner magnetometer. Some specimens were subjected to alternating field (AF) demagnetization in steps, (typically 14) up to the maximum available field of 100 mT using 2 and 3 axis instruments. Other specimens were subjected to thermal demagnetization, using a Schonstedt TSD-1 furnace, up to maximum temperatures of 700°C. In the thermal demagnetization experiments, the number of separate heatings and coolings followed by remanence measurement, varied between eight and 13 depending upon the ultimate unblocking temperatures. A few specimens were subjected to AF demagnetization followed by thermal demagnetization. No 'blanket cleaning' procedures were used. Characteristic directions, and any other components manifested by linear segments on orthogonal demagnetization diagrams (Zijderveld, 1967), were estimated using the Principal Component Analysis computer program of Kirschvink (1980) as modified by Halls (1986). This modification generates all possible linear segments defined by three or more data points and a mean angular deviation (MAD) of fit $\leqslant 5°$. Some of the characteristic directions in specimens from sites 11, 15, 17 and 21 were defined with lower precision and for these specimens the MAD angles for acceptance were relaxed to 10°. Specimen–sample characteristic directions were vectorially averaged to produce the site-means which are given in Table I.

Paleomagnetic results

Useful paleomagnetic results were obtained from samples from 23 of the 25 sampling sites. Although individual sample directions are defined with high precision in site 11, they show a wide range in NRM intensities and no within-site directional agreement. The remanence in site 11 samples is believed to have been affected by lightning and the results from this site are discarded from further analysis. Site 18 occurs in a small portion of the Mamainse Point Formation outcrop area which Annells (1974) shows as having an anomalous northerly dip. A definitive structural attitude could not be determined at the collecting locality. Application of a steep northerly dip correction brings the site direction toward, but not within, the cluster of other Mamainse Point Formation directions (Fig. 3). Its direction can only be made to coincide with the other Mamainse Point Formation site-means by assuming that the rocks at the collecting locality are overturned for which there is no direct observational evidence. Furthermore, the NRM intensities of the rocks at this site are one to two orders of magnitude less than those at other Mamainse Point Formation localities possibly suggesting that these rocks are hydrothermally altered. As a consequence the results from site 18 are also discarded from further analysis.

The paleomagnetic results from the remaining 23 sites can be interpreted in a straightforward way. In sites from the Mamainse Point Formation and from all members of the Michipicoten Island Formation with the exception of the Davieaux Island Rhyolite the remanence consists of a single component which decays linearly to the origin under alternating field (AF) or thermal demagnetization. In Fig. 4 are shown typical results.The results from

TABLE I

Characteristic site mean directions

Rock unit	Site No.	N	R	k	α_{95}	In situ		θ	ϕ	Q	Tilt cor.	
						D	I				D	I
MP	19	4	3.996	809	3.2	337	04	116	51S	A	321	33
MP	18	6	5.808	26	13.4	207	−24	270	80N	C	217	47
MP	7	7	6.886	52	8.4	283	19	020	42E	A	277	61
MP	8	6	5.847	33	11.9	317	16	110	42S	A	299	30
MP	22	6	5.987	399	3.4	308	39	100	40S	A	270	46
MP	10	6	5.990	508	3.0	322	30	110	32S	A	300	41
MP	9	5	4.993	537	3.3	322	31	090	35S	A	296	54
MP	23	3	2.999	2044	2.7	318	18	090	33S	A	303	40
MP	KM	6	5.927	68	8.2	321	29			A	299	51
FIP	20	4	3.892	28	17.7	286	−15	070	20S	C	288	−03
FIP	21	4	3.979	140	7.8	309	−23	090	51S	C	313	12
SIP	6	5	4.888	36	13.0	303	08	080	25S	C	297	25
CM	11	4	1.736	1	-	329	18	090	25S	C	321	39
CM	1	6	5.772	22	14.6	309	09	090	19S	A	304	20
CM	24	5	4.903	41	12.1	311	18	110	30S	A	299	26
CM	12	4	3.800	15	24.5	281	12	070	19S	C	276	21
CM	25	4	3.990	313	5.2	311	−10	070	20S	B	311	08
CLM	3	5	4.991	432	3.7	297	−08	077	20S	A	297	06
CLM	13	4	3.970	101	9.2	301	23	065	17S	C	296	37
QHM	2	4	3.980	152	7.5	314	24	100	22S	A	303	35
QHM	4	5	4.982	228	5.1	318	31	080	14S	A	312	43
SSM	14	5	4.979	188	5.6	290	20	095	19S	C	283	24
SSM	5	5	4.982	224	5.1	295	12	070	30S	B	286	32
SSM	16	4	3.956	68	11.2	286	07	080	16S	C	283	13
DIM	15	3	2.929	28	23.7	290	−09	070	21S	C	291	05
DIM	17	4	3.981	160	7.3	294	23	077	22S	B	283	34

Rock unit designations are as follows: MP, Mamainse Point Formation of Michipicoten Island; FIP, Intrusive rocks of first intrusive phase; SIP, Intrusive rocks of second intrusive phase; CM, Cuesta Member of Michipicoten Island Formation; CLM, Channel Lake Member of Michipicoten Island Formation; QHM, Quebec Harbour Member of Michipicoten Island Formation; SSM, South Shore Member of Michipicoten Island Formation; DIM, Davieaux Island Member of Michipicoten Island Formation.

Locations of sites are shown in Fig. 2 with the exception of site 'KM'. 'KM' consists of five block samples collected between sites 8 and 6. Structural attitudes used in making the restoration to paleohorizontal for these block samples are within the range tabulated for sites 8 through 23. N is the number of specimens used in the analysis. R, k and α_{95} are the statistical parameters of Fisher (1953). θ and ϕ are the strike and dip of the sampled flows. Q is a quality factor describing the confidence of the flow attitude. The grades for the Q rating are as follows: A, readily measured in outcrop and evidence that the attitude is constant over tens of meters; B, measured in outcrop, but attitude is seen to vary along flow surfaces; C, attitude indeterminate in outcrop and attitude is estimated from regional attitudes or from attitudes of enclosing rocks when site is in an intrusive unit.

thermal demagnetization experiments indicate that the single component of remanence present in these rocks is carried by both magnetite and hematite. In only a few samples from the collection is a spurious remanence component present which can be removed at the lowest fields and temperatures of demagnetization.

The remanence in the samples from the Davieaux Island Rhyolite and from the intrusive units is not as stable as that in the other extrusive units. The intensities of NRM are lower and their orthogonal demagnetization diagrams have a more jagged character than those depicted in Fig. 4. Nevertheless, in spite of the

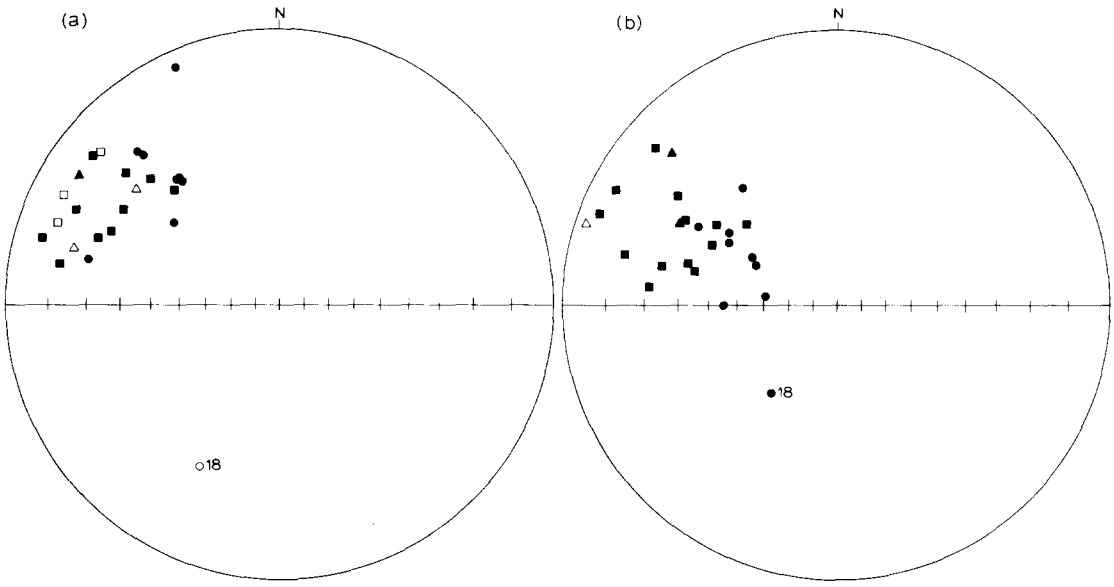


Fig. 3. Stereographic projections showing the characteristic directions from Mamainse Point sites (circles), intrusive rock sites (triangles) and Michipicoten Island Formation rock sites (squares). Closed (open) symbols are on the lower (upper) hemisphere. In part (a) the directions are referenced to the present horizontal whereas in (b) the directions are referenced to the estimated paleohorizontal. Site 18 is from a fault bounded block (see Fig. 2) having anomalous northerly dip.

characteristic directions being defined by larger MAD angles, within-site grouping of directions in the intrusive units and the Davieaux Island Rhyolite is still quite respectable (Table I).

The site-mean directional data of Table I have been grouped to provide rock unit-directional means for the Mamainse Point Formation, the Michipicoten Island Formation and for the intrusive units (Table II, Fig. 3). Although application of the fold test (Graham, 1949) is not strictly applicable, since the sequence is crudely homoclinal, dispersion for each unit decreases upon application of the tilt correction. Because the time of tilting is poorly constrained, however, this fact cannot be used to argue that the remanence is primary. The principal line of evidence that the remanence is primary remains with the observation that the remanence is single component and is carried by both magnetite and hematite. This argument is made more powerful by the fact that primary volcanic glass is still preserved in the Michipicoten Island Formation (Annells, 1974). The preservation of this original glass indicates a simple cooling history following original crystallization and magnetization.

Geochronology

A quartz–feldspar porphyry intrusive rock from the eastern end of the island (site 20, Fig. 2) was selected as the unit most likely to have zircon and a 40 kg sample was collected. Rock crushing and mineral separation were carried out under stringently controlled clean conditions to minimize any possibility of contamination. Mineral separation was carried out using a Wilfley table, Frantz magnetic separator and heavy liquids. The zircons were divided into different magnetic fractions. Only the least magnetic fractions were used for hand picking.

The rock contains abundant, clear euhedral zircon, with well developed primary crystal faces. Most grains are fairly equant. Previous experience with Keweenawan felsic rocks has shown that one of the major problems in obtaining precise ages is the presence of inher-

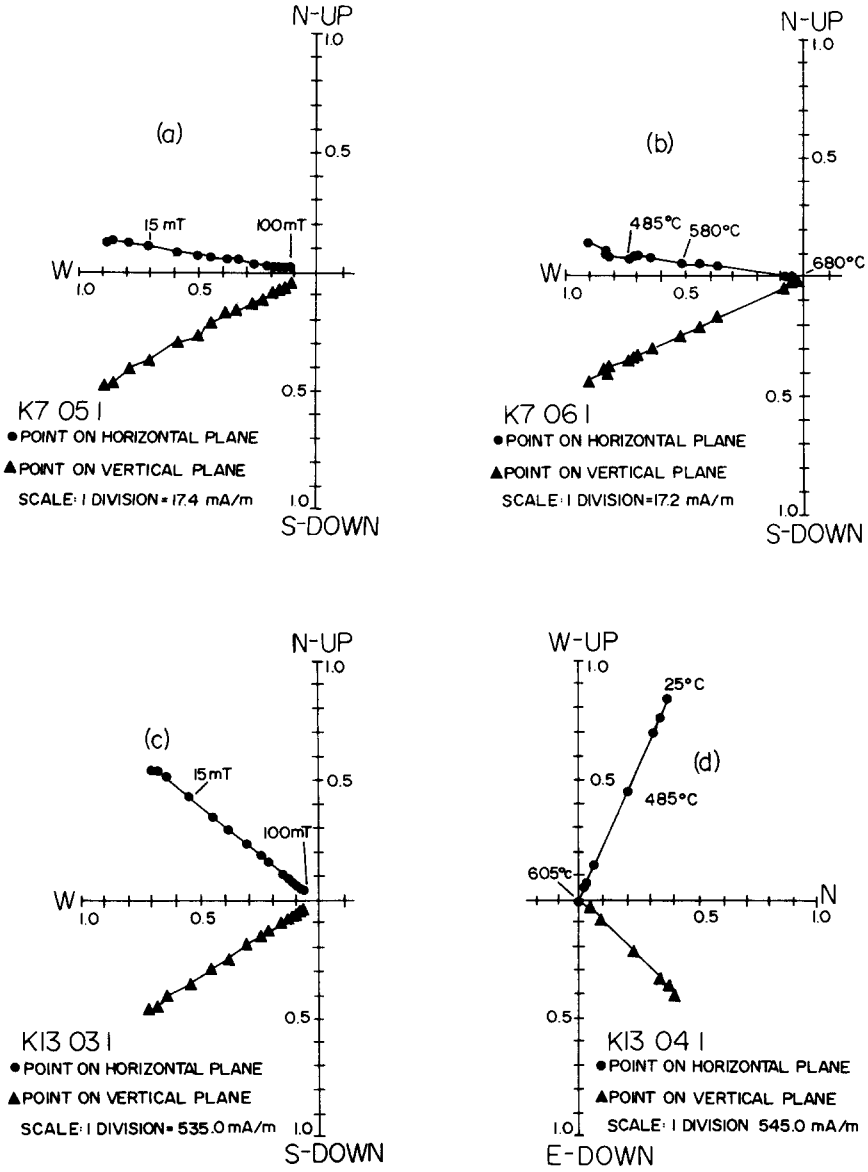


Fig. 4. Orthogonal vector diagrams from companion samples from site 7 in the Mamainse Point Formation basalts (a, b) and from companion samples from site 13 in the Channel Lake andesites (c, d). In all parts of the figure, vectors are referenced to the present horizontal. Note the similarity of characteristic directions whether obtained by AF or thermal demagnetization. The characteristic direction at site 7 is carried by both magnetite and hematite, whereas at site 13 hematite appears to be absent.

ited components in the zircon populations derived from melting of older crust (Davis and Sutcliffe, 1985). This is present in the porphyry as shown by core-overgrowth relationships in some grains. The cores are generally high in uranium and are metamict, which has resulted in fracturing of the lower uranium overgrowth material (Fig. 5).

Inherited components are most visible when there is a large uranium contrast between core and overgrowth and hence a large contrast in the degree of metamictization. If the older cores

TABLE II

Formational mean directions and poles

In situ						Tilt corrected						Paleopole		
N	*R*	*k*	*α*95	*D*°	*I*°	*N*	*R*	*k*	*α*95°	*D*°	*I*°	Long°	Lat°	A95°
Mamainse Point Formation														
8	7.660	20.6	12.5	316.3	+23.9	8	7.753	28.4	10.9	297.3	+45.5	186E	38N	11
Michipicoten Island Formation														
13	12.435	21.2	9.2	299.4	+12.1	13	12.516	24.8	8.5	293.8	+23.8	175E	26N	7
Intrusive Units														
3	2.853	13.6	34.7	298.5	−07.7	3	2.881	16.8	31.1	298.2	+13.4	166E	24N	22

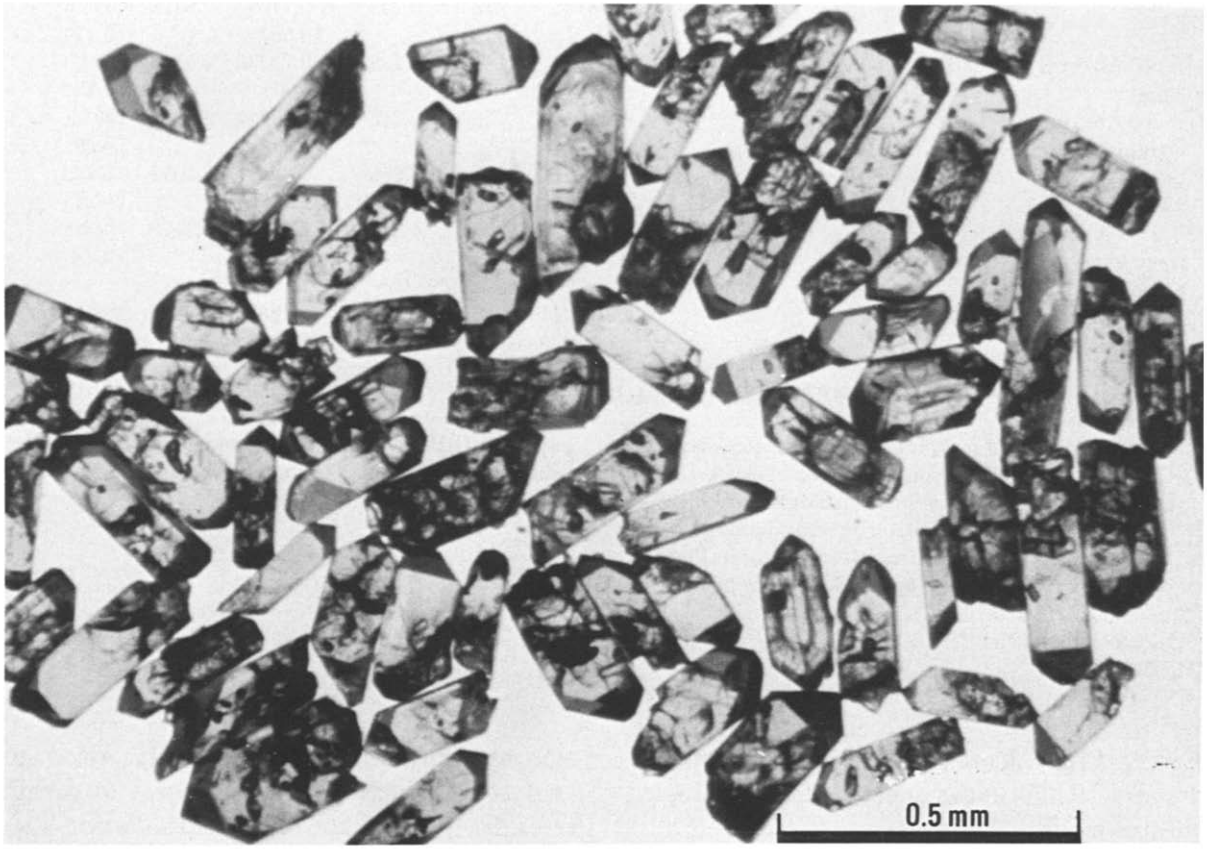


Fig. 5. Zircons with visible cores. Expansion of the more metamict cores has resulted in cracking of the mantle material.

are similar in uranium content to the overgrowths, however, they may not be readily visible under the microscope. During hand picking of the first five fractions (Table III), attention was paid to eliminate any grains with visible cores. These were selected to assess whether inheritance could be eliminated by visual selection of grains. For the first three fractions, only clear, crack-free grains were picked, and these were abraded to eliminate most secondary lead loss (Krogh, 1982). Dissolution and chemistry were carried out using standard techniques

TABLE III

Isotopic data for zircons from the Michipicoten Island porphyry

Zircon fractions	Weight (mg)	U (ppm)	(a) Common Pb (pg)	(b) $^{207}Pb/^{206}Pb$	(b) $^{207}Pb/^{204}Pb$	(c) $^{208}Pb/^{206}Pb$	(d) $^{206}Pb/^{238}U$	$^{207}Pb/^{235}U$	(e) $^{207}Pb/^{206}Pb$ Age (Ma)
1. +200 Ab	0.254	98	27	0.07662	1173	0.253	0.18259 (0.50)	1.9055	1087.0 (0.14)
2. −200 Ab	0.226	130	31	0.07655	1185	0.248	0.18286 (0.50)	1.9069	1085.2 (0.11)
3. +200 Ab	0.040	93	9	0.07587	26845	0.270	0.18240 (0.60)	1.9069	1090.3 (0.14)
4. +200	0.545	99	241	0.08123	216.7	0.262	0.17774 (0.50)	1.8602	1092.4 (0.10)
5. −200	0.315	132	188	0.08145	213.2	0.259	0.17556 (0.50)	1.8403	1095.6 (0.11)
6. +200 Ab cores 3 grains	0.017	123	26	0.09254	151.7	0.138	0.21736 (1.00)	2.5165	1292.0 (0.22)
7. +200 Ab cores 6 grains	0.005	192	24	0.09802	78.76	0.213	0.20072 (1.00)	2.2262	1207.7 (0.39)
8. +200 Ab cores 5 grains	0.008	167	19	0.08801	137.2	0.255	0.19312 (1.00)	2.1018	1170.3 (0.24)
9. +200 Ab mantle fragments, 6 grains	0.003	177	19	0.09907	60.01	0.210	0.18398 (1.00)	1.9178	1084.4 (0.48)
10. +200 Ab clear tips	0.050	94	21	0.07894	348.4	0.237	0.18329 (0.60)	1.9135	1087.5 (0.12)

+200 and −200 means grains which are larger and smaller, respectively, than 200 mesh size (about 120 microns).
Ab = abraded fraction.
(a) Corrected for 0.117 mole fraction of common lead in the ^{205}Pb spike, includes blank + zircon common lead.
(b) Corrected for fractionation and blank: U blank = 3.8pg (large column), 0.5pg (small column); Pb blank = 8.2pg (both columns); Pb and U fractionation correction = 0.13%/amu.
(c) Corrected for fractionation, blank and common Pb.
(d) Number in parentheses refers to % 2σ error in Pb/U.
(e) Number in parentheses refers to % 2σ error in $^{207}Pb/^{206}Pb$ ratio.
$\lambda^{238}U = 0.15513 \cdot 10^{-9}\ a^{-1}$.
$\lambda^{235}U = 0.98485 \cdot 10^{-9}\ a^{-1}$.
$^{238}U/^{235}U = 137.88$.

(Krogh, 1973). Isotopic analyses were carried out on a VG354 mass spectrometer in single collector mode.

All age errors are quoted as 95% confidence levels. Errors for the data points are estimates of internal reproducibility, including the effect of common lead correction. Data regression and age errors were calculated using the program of Davis (1982). The results are shown in Fig. 6 and in Table III. The first three abraded fractions consisted of coarse grains, fine grains and elongate or flat grains, respectively. Although analyses of fractions 1 and 2 agree to within error, fraction 3 is too old outside of error, suggesting that a small amount of inherited component is present (Table III). In addition, this fraction appears to have a small amount of secondary lead loss perhaps due to the eccentric shape of the grains which makes abrasion of the surfaces difficult. Fractions 4 and 5, which were not abraded, display secondary lead loss. Their $^{207}Pb/^{206}Pb$ ages are older than the ages of the

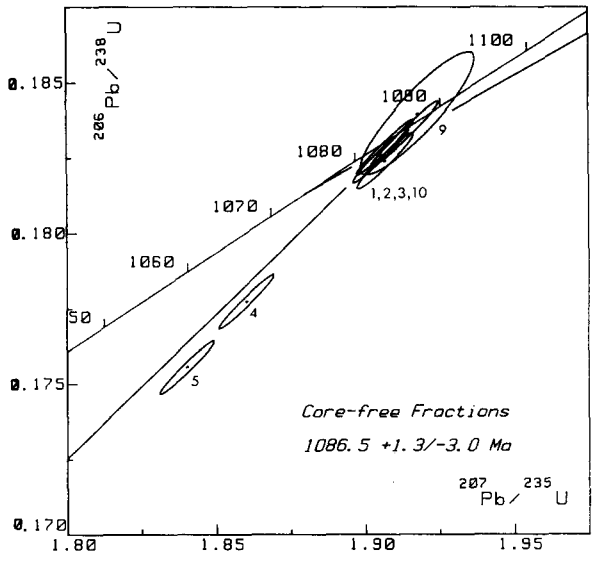


Fig. 6. U-Pb analyses of zircon fractions which were free of visible cores. Fractions 4 and 5 were not abraded. The lower discordia line is defined by the youngest fractions (1, 2, 9 and 10) with a lower intercept of 0 Ma. The upper line is a mixing line defined by the core fractions as shown in Fig. 7.

abraded fractions discussed above, showing that they contain inherited material. As shown in Fig. 6, all five fractions would define a lead loss – inheritance line to the right of zero. Since there is no reason for lead loss to be correlated with inheritance, this line must be fortuitous and we cannot use the upper intercept defined by these fractions to calculate a geologically meaningful age.

To isolate this inherited component, we picked grains having clear, relatively low uranium, crack-free cores. The overgrowths were split off using tweezers under the microscope, and the cores were abraded to eliminate secondary lead loss and any remaining rim material. Since grains having these characteristics are quite rare in the sample, it was necessary to analyze very small fractions consisting of only a few grains. The cores would also be expected to have lost some lead due to thermal diffusion during magmatic emplacement of the porphyry. Therefore, it was necessary to analyze several fractions to define a mixing line between the age of porphyry emplacement and the age of the inherited component. Three fractions were analyzed (fractions 6, 7 and 8). Because of their small size, they were dissolved by HF vapour on thin teflon film. Since the dissolution container was disposable this eliminated the possibility of cross contamination (memory). Isotopic analyses had to be carried out using a Daly detector. A lead standard was run during the course of the analyses to precisely monitor peak shape and detector mass fractionation.

To obtain fractions most likely to be free of inheritance, we separated and abraded a very small fraction of crack-free rim fragments (fraction 9). Since the inherited components in the grains from which these fragments were derived were recognized and separated, we can be reasonably sure that this fraction contains no inheritance. In addition, fraction 10 was produced by breaking the tips off long, clear grains and strongly abrading them.

The results of all isotopic analyses are plotted in Fig. 7. Fractions 6 to 9 have larger errors because of the Daly detector analyses. The overgrowth fragments (fraction 9) have the youngest age (1084.4 ± 10 Ma), however, this overlaps with the ages of the four larger abraded fractions due to the large error. The three core fractions combined with the other abraded fractions (excluding 3, which may have some secondary lead loss), define a line with a reasonable probability of fit (43%). It has an upper intercept of 1403 ± 50 Ma and a lower intercept of 1074 ± 10 Ma. The lower intercept age has a rather large error due to the small angle which the mixing line makes with concordia. For this reason also, it is extremely sensitive to any small degree of secondary lead loss affecting the large abraded fractions. We would therefore consider it to be only a minimum estimate for the age of emplacement of the porphyry. Assuming that the four youngest fractions (1, 2, 9 and 10) are free of inheritance, an age can be calculated from a best fit line through these data points and the origin. This gives an age of 1086.5 ± 1.3

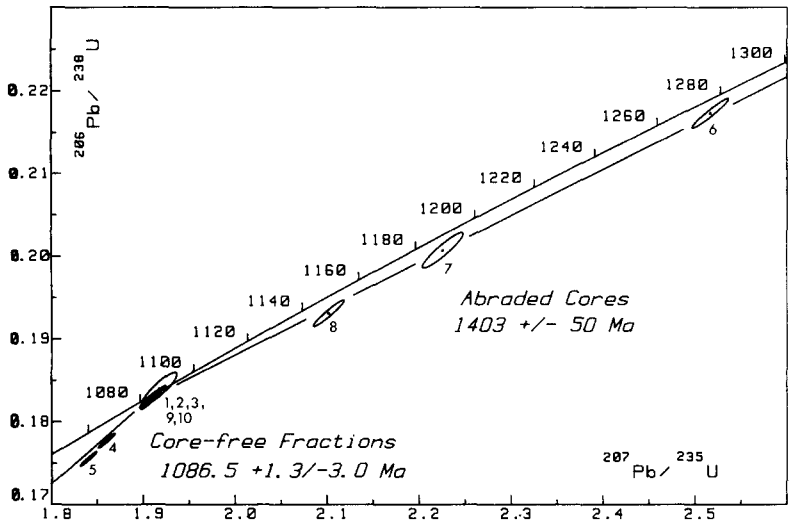


Fig. 7. U-Pb analyses of all zircon fractions. The abraded cores (fractions 6, 7 and 8) define a mixing line with an upper intercept of 1403 ± 50 Ma.

Ma. The line through the origin has an excellent probability of fit (55%), which indicates scatter within experimental error. This tight grouping would not be likely if significant amounts of inheritance were present. However, considering the possibility that some of the fractions may have minor inheritance, which would bias the average age upwards, the lower age error limit should be increased to easily encompass the $^{207}Pb/^{206}Pb$ ages of all four data points. We therefore consider 1086.5 + 1.3/−3.0 Ma to be a reasonably accurate estimate of the age.

Discussion

The upper intercept age of 1403 Ma ± 50 Ma for the abraded fractions represents the age of zircons in the source material which contributed, after melting, to the magma emplaced at Michipicoten Island. It is quite different from Archean upper intercept ages previously obtained from rhyolites of the Osler Group north of Lake Superior (Davis and Sutcliffe, 1985). A large number of mid-Proterozoic anorogenic granite plutons were emplaced into the stable craton of North America in a broad belt extending from California to Labrador. The peak of their emplacement was from about 1400 to 1500 Ma (Anderson, 1983). The presence of such a body under the Keweenawan rift may be the reason for a concentration of felsic igneous activity at this location. These bodies were emplaced as relatively dry magmas. Anderson (1983) estimates their water content to have been generally < 1.5%. This may help to explain why inherited zircons, which should dissolve rapidly in a hydrous melt, but are relatively stable in a dry one, have survived to contaminate the Michipicoten porphyry (Harrison and Watson, 1983).

The pole positions for the Michipicoten Island Formation and the Mamainse Point Formation as exposed on Michipicoten Island are plotted in Fig. 8. The Michipicoten Island poles (Fig. 8) are still north of the Nonesuch Shale pole (Fig. 1) but this is now in accordance with the age data. The pole from the intrusive units, although recorded in Table II, is not plotted as it is based on only three sites for which the structural correction is uncertain. The best estimate of the age of emplacement of the porphyry, 1086.5 + 1.3/−3.0 Ma serves to give a younger age limit to the MP pole and an older

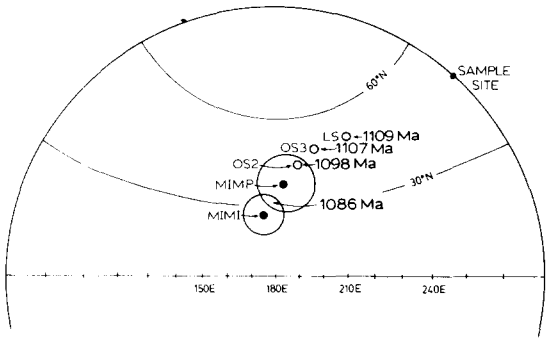


Fig. 8. Paleomagnetic poles from Keweenawan units dated by modern U-Pb methods. Open circles are poles of reversed polarity whereas closed circles are poles of normal polarity. LS = mean Logan Sills (Halls and Pesonen, 1982); OS3 = lower part of Osler Reversed Sequence (Palmer, 1970); OS2 = upper part of Osler Reversed Sequence (Halls, 1974); MIMP = Michipicoten Island, Mamainse Point Formation (this study); MIMI = Michipicoten Island, Michipicoten Island Formation (this study). Ages correlated with poles LS, OS3 and OS2 are from Davis and Sutcliffe (1985). The age of 1086 Ma, recovered from the site 20 porphyry, post-dates the Mamainse Point Formation and pre-dates the Michipicoten Island Formation. The 1107 Ma age for pole OS3 may be too old inasmuch as the age is determined from a porphyry at the base of the Osler Group, whereas the pole is determined from basalt flows stratigraphically above the dated porphyry.

limit to the MI pole. Several reversely magnetized Keweenawan units from the north shore of Lake Superior have been dated by high precision U-Pb methods (Davis and Sutcliffe, 1985). They are the Logan Sills at $1108.8+4/-2$ Ma; a porphyry at the base of the Osler Group at $1107.5+4/-2$ Ma and a rhyolite near the top of the reversed portion of the Osler Group at 1097.6 ± 3.7 Ma. The Osler Group has been studied paleomagnetically by both Palmer (1970) and Halls (1974). Palmer's study involved flows from the lower part of the group whereas Hall's study involved flows from the upper part of the group including flows above the reversed to normal polarity transition. The poles are plotted in Fig. 8 with the associated U-Pb ages. The 1107 Ma porphyry lies beneath basalts of the Osler Group and may be slightly older than the OS3 pole. The low level of secondary alteration in these Keweenawan rocks as a whole and its virtual absence in the Michipicoten Island rocks, as manifested by the survival of primary volcanic glass, indicates a simple and relatively fast cooling history. Therefore, in these rocks isotopic closure and magnetic blocking were essentially synchronous. When Figs. 1 and 8 are compared it can be seen that the age sequence of poles as determined stratigraphically is in complete agreement with their assigned absolute ages. Assuming that no displacement along a parallel of paleolatitude is involved, these can be used to calculate a minimum rate of continental drift. The most precise value is that calculated from the two end points, the Logan Sills and the Michipicoten Island porphyry. This gives 14 cm a^{-1}, a value not untypical of Phanerozoic plate velocities during times of active spreading (Patriat and Achache, 1984). The time span of 22 Ma between the ages of the Logan Sills and the Michipicoten Island porphyry probably encompasses the bulk of Keweenawan volcanism, since the Logan Sills define one of the earliest reversed pole positions on the Keweenawan apparent polar wander track and most of the normally magnetized poles for igneous rocks, including those from Michipicoten Island, cluster tightly near the middle of the track (Halls and Pesonen, 1982; Fig. 1, this paper). The time of the magnetic reversal is constrained to be between the age of the rhyolite from the upper part of the Osler Group, 1097.6 ± 3.7 Ma and the Michipicoten Island porphyry at $1086.5+1.3/-3.0$ Ma. If suitable material can be found at either Mamainse Point, Gargantua Point or the north shore of Minnesota, a more precise age for the time of the reversed to normal polarity transition may be obtained as this reversal is recorded at all three localities.

Conclusions

A small polar shift is recorded from the volcanic rocks of Michipicoten Island. The pole from the Mamainse Point Formation lies to the north of that derived from the overlying Michipicoten Island Formation. The apparent

polar movement from north to south with time is in accord with other Keweenawan paleomagnetic observations and no back-tracking of the pole is required.

The age of the normally magnetized Michipicoten Island porphyry is $1086.5 + 1.3/-3.0$ Ma. The upper part of the Mamainse Point Formation and its normal pole at 188°E, 38°N is older than the porphyry but younger than the age of the upper Osler rhyolite, 1097.6 ± 3.6 Ma.

Comparison of the ages and the paleomagnetic pole positions of both normally and reversely magnetized Keweenawan rocks leads to an estimate for the minimum polar wander rate of 14 cm a^{-1} about 1.1 Ga ago. The drift rate may have been greater if longitudinal displacements of the North American plate were taking place.

The time span for the bulk of Keweenawan igneous activity is about 22 Ma, between the age of the Logan sills and the Michipicoten Island porphyry. The time of magnetic reversal is between $1086.5 + 1.3/-3.0$ and 1097.6 ± 3.6 Ma, the age of the upper Osler rhyolite.

One of the porphyries on Michipicoten Island contains inherited zircon from a source having an age of 1403 ± 50 Ma. The porphyry was therefore at least partly derived by partial melting of an older felsic source which probably represents an anorogenic pluton. This suggests that the 'eastern granite–rhyolite province' of Bickford et al. (1986) extends north into the Superior craton.

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